Load transfer is not a binary payload switch

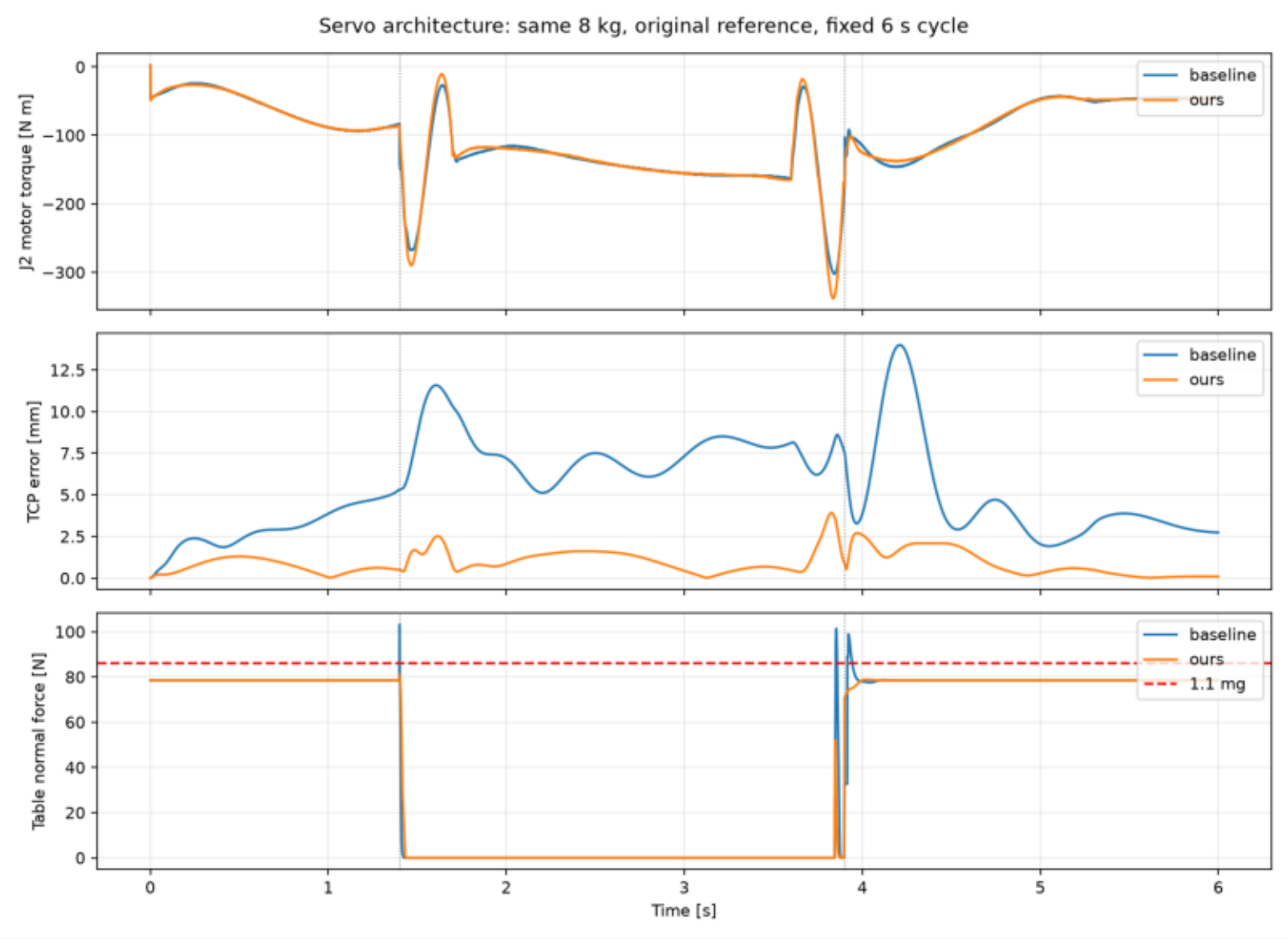
Individual advanced manipulation study: M1013 palletizing, A servo vs B torque.

Result: elastic feedforward + continuous load transition reduce torque steps and TCP error.

Wrench-consistent classification removes official no-collision false alarms.

Not solved: <15 N transport sensitivity, sustained pre-release unloading, universal robustness or hard real time.

Problem definition: baseline A



Idea 1: elastic motor reference and feedforward

theta_d = q_d + K^-1 tau_J,d; tau_J,d = tau_0 + lambda_hat (tau_box - tau_0).

tau_ff = B theta_ddot + tau_J,d + viscous + tanh Coulomb friction.

One-sample trajectory advance; bounded/filtered elastic derivatives.

A retains the fixed cascaded drive. B adds low-bandwidth motor PD and small JTS torque feedback.

B2 uses an inertia-shaping interpretation, not a demonstrated sampled-data passivity theorem.

Idea 2: table-consistent wrench and blind thresholds

p = M(qhat) vhat; r = K_O [p - integral(tau_s - bias_hat - beta + r)dt].

w_hat = J_bottom^-T r; subtract feasible upward support with COP inside the rotated footprint.

Support allowed only with grip, positive box mass and taught table-height band.

Residual horizontal/negative-normal/COP/torsion/out-of-height evidence + grip-off transient channel.

Symmetric amplitudes and persistence fitted on no-collision training only; disjoint holdout unused for fitting.

Idea 3: shared estimator, separate contributions

Standalone files embed identical estimator code; optional self.ctrl connection, no runtime helper import.

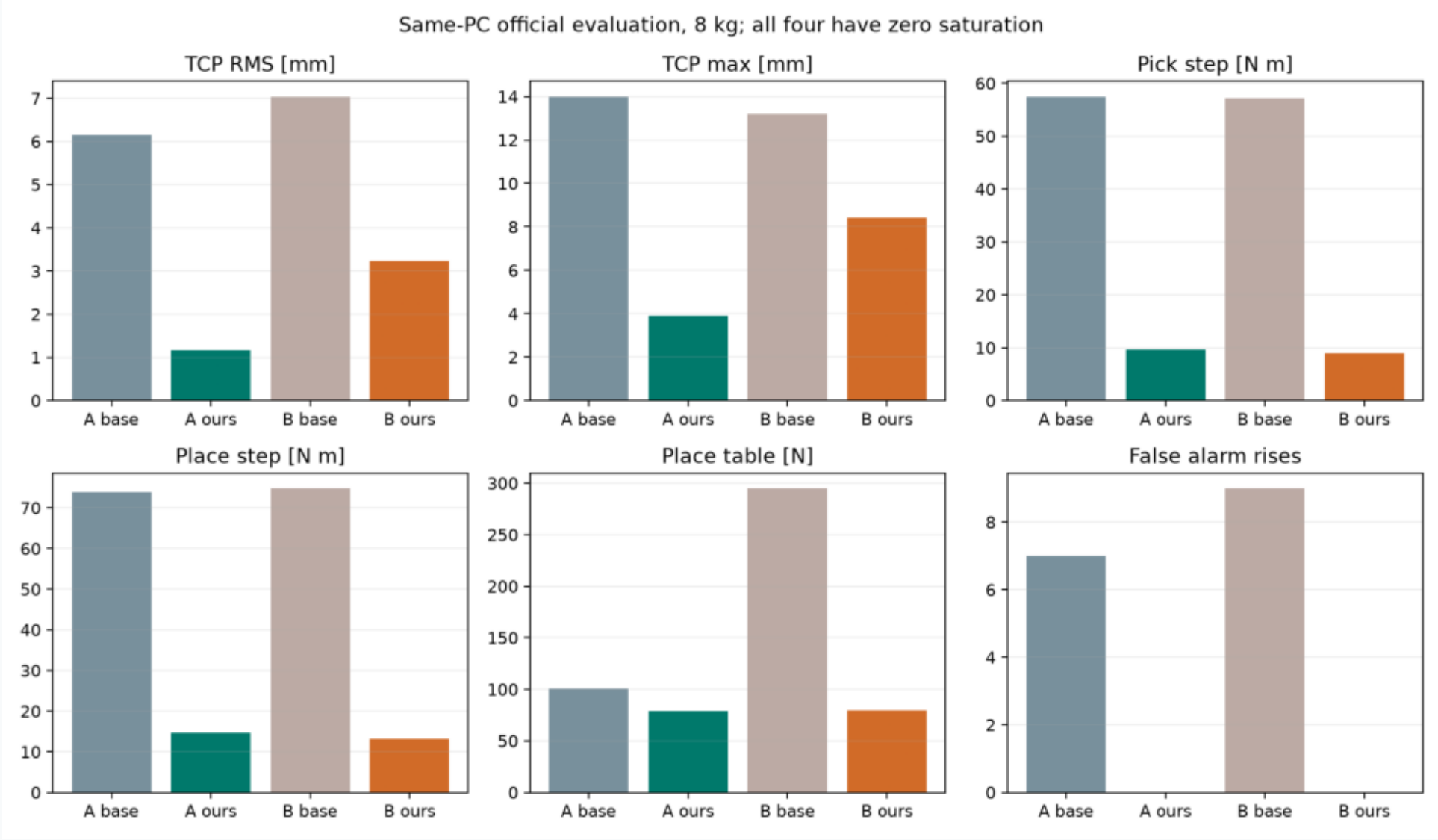
Detector executes first; timestamp caching gives one estimator advance per sample.

Same physical and detection outputs: shared 6000 vs independent 12000 advances.

10% bounded support fusion is a separate algorithmic contribution; sharing itself saves compute only.

Aggressive early-settle candidates were rejected. Sustained handover before release remains unmet.

Official evaluation: baseline vs final



Ablation: changes that helped and changes that did not

A baseline: 6.14/14.00 mm RMS/max; pick/place step 57.5/73.8 N m.

C1 alone: 1.17/3.37 mm but 107.7/106.6 N m. Elastic tracking alone worsens edge steps.

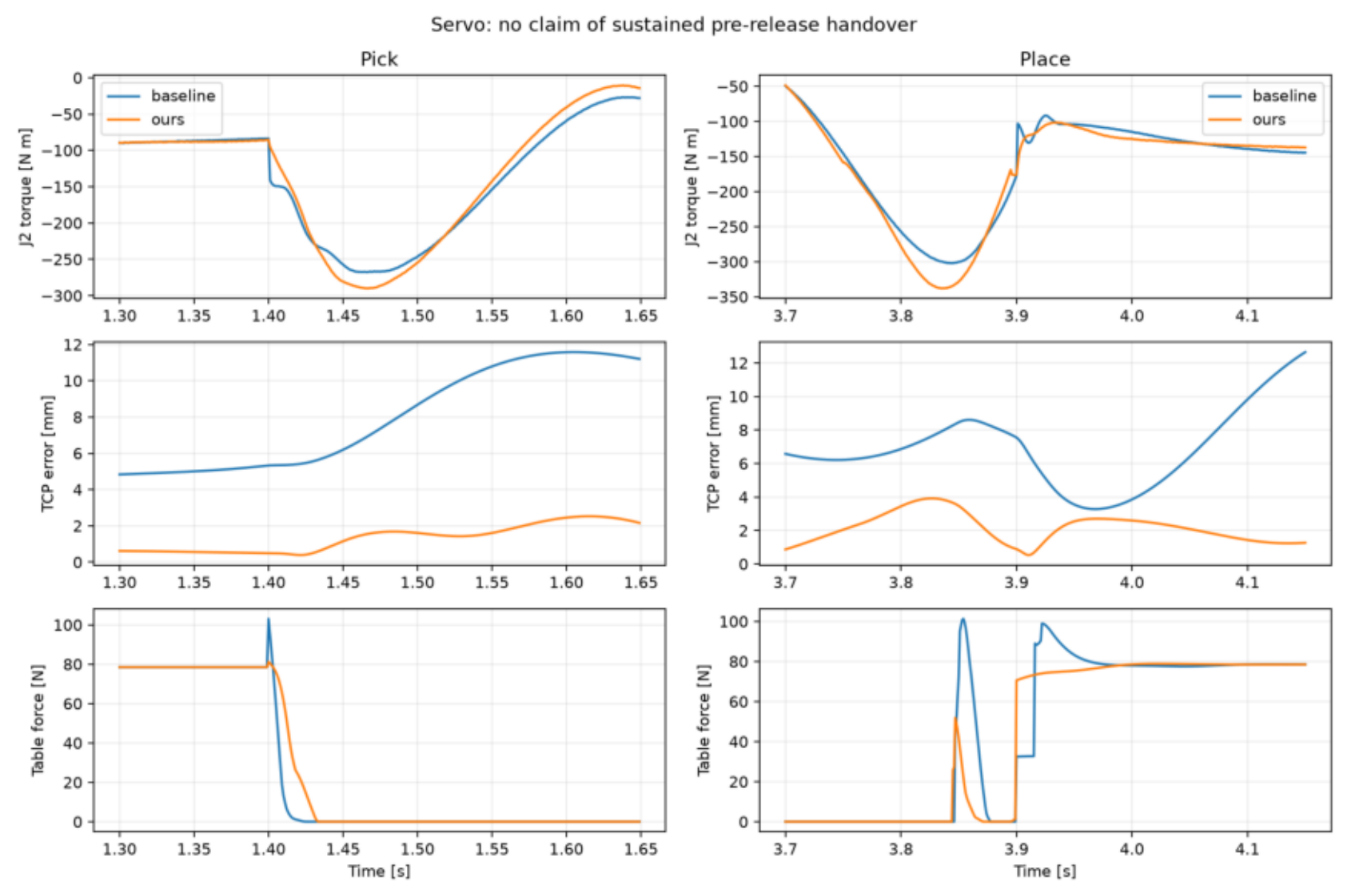
C1+C2: 1.20/3.45 mm, 9.7/17.7 N m. Continuous load FF is the main step reduction.

B baseline: 7.04/13.18 mm, 57.2/74.8 N m. B1 alone: 2.24/4.45 mm, 97.9/96.7 N m.

V reduces pickup Fz peak but not overall oracle RMSE; bias/wrench logic and blind thresholds remove false alarms.

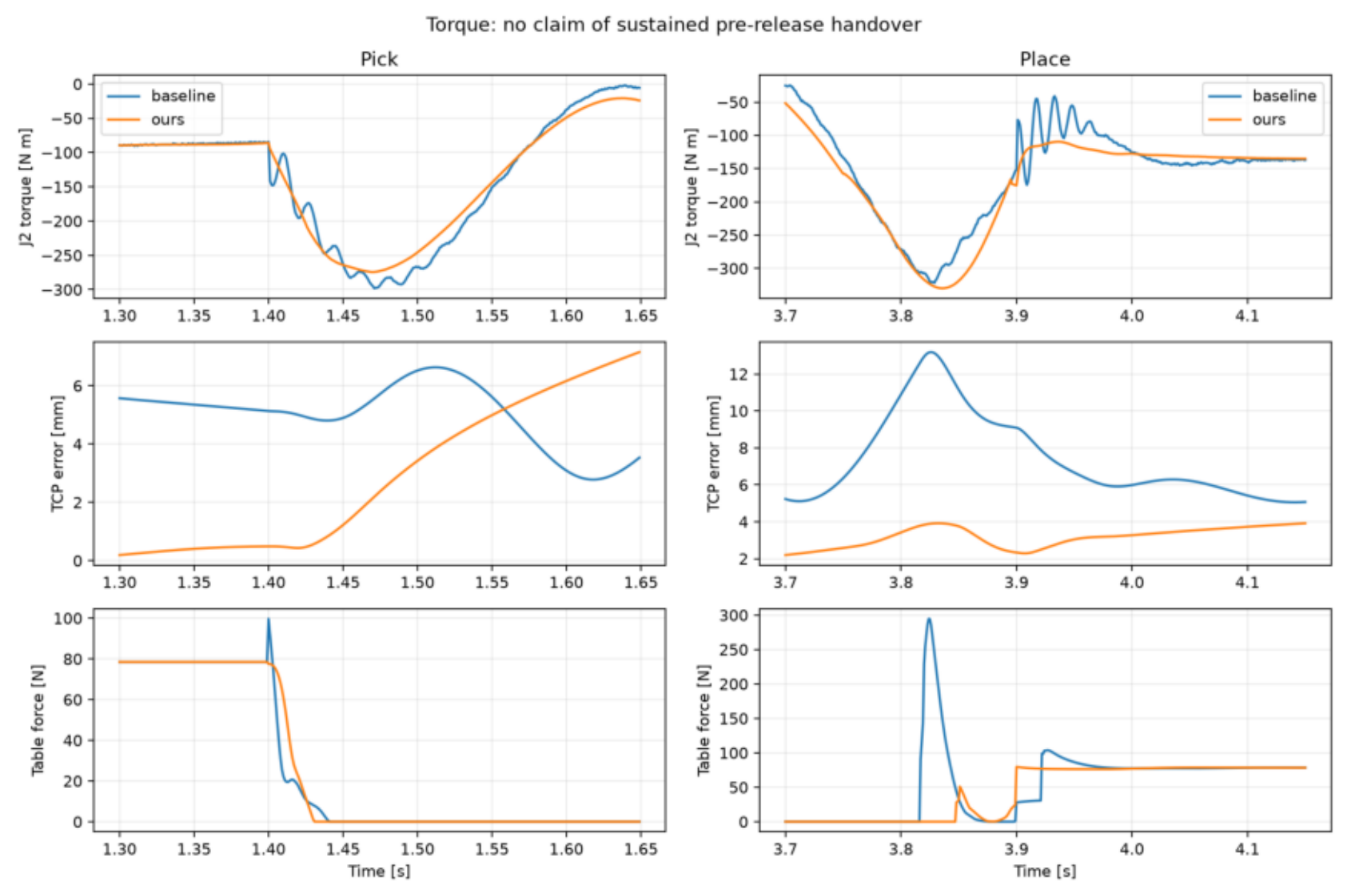
Historical stages, final support fusion and computation reuse are separated in the result tables.

A transfer detail: original-reference error cost

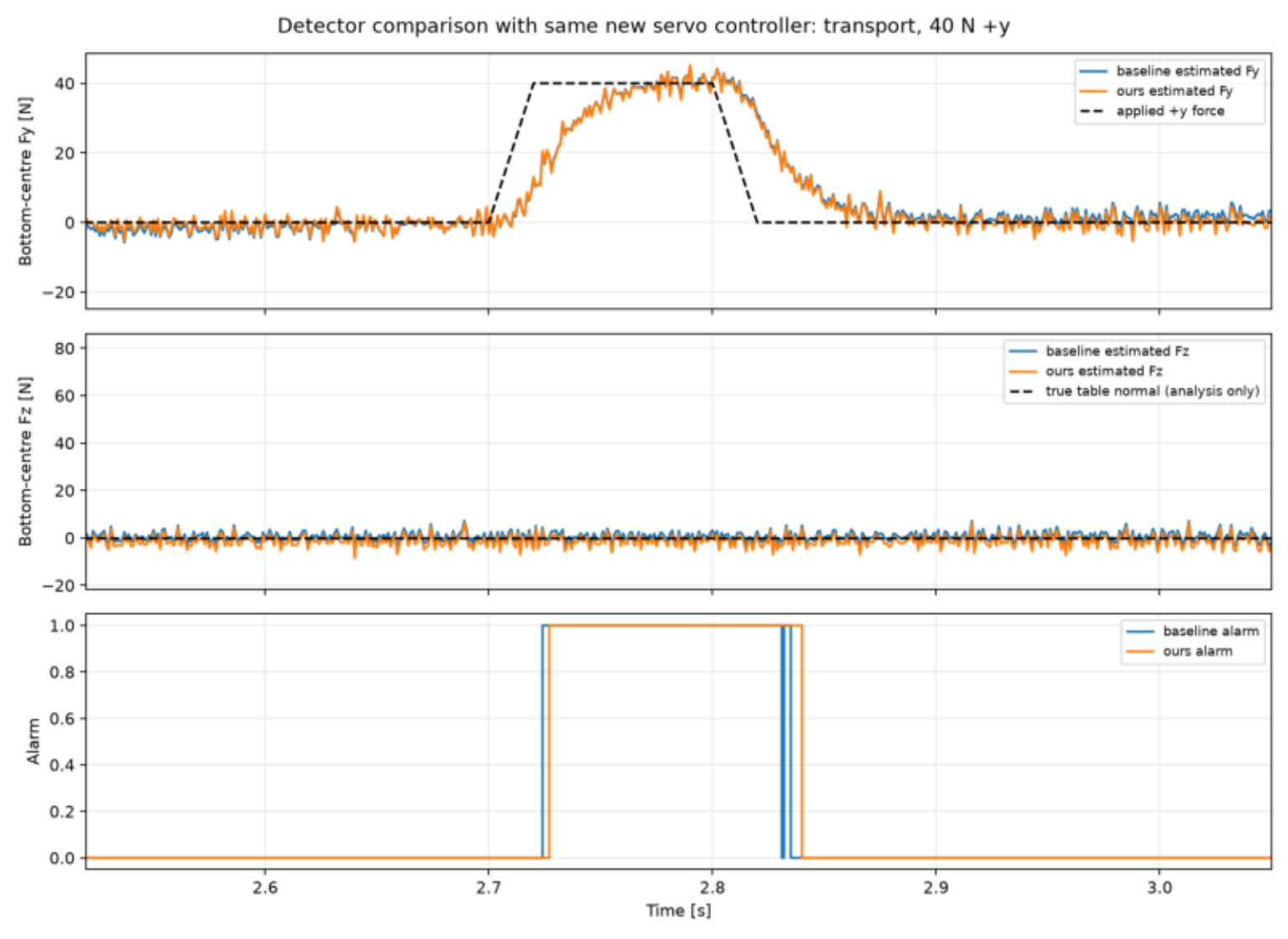


Simulation study | fixed 6 s cycle | official scores separated from analysis

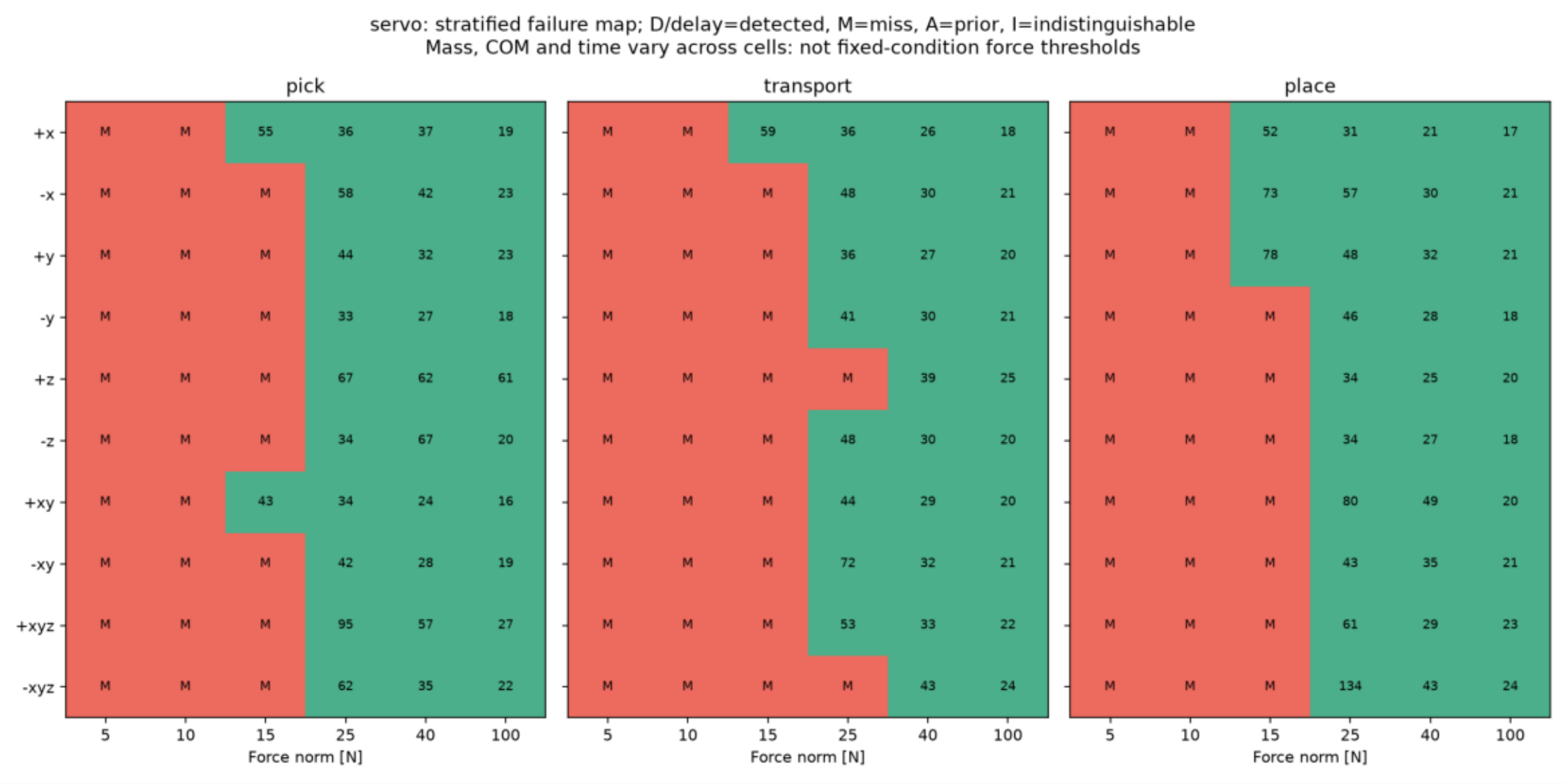
B transfer detail: freedom is not free robustness



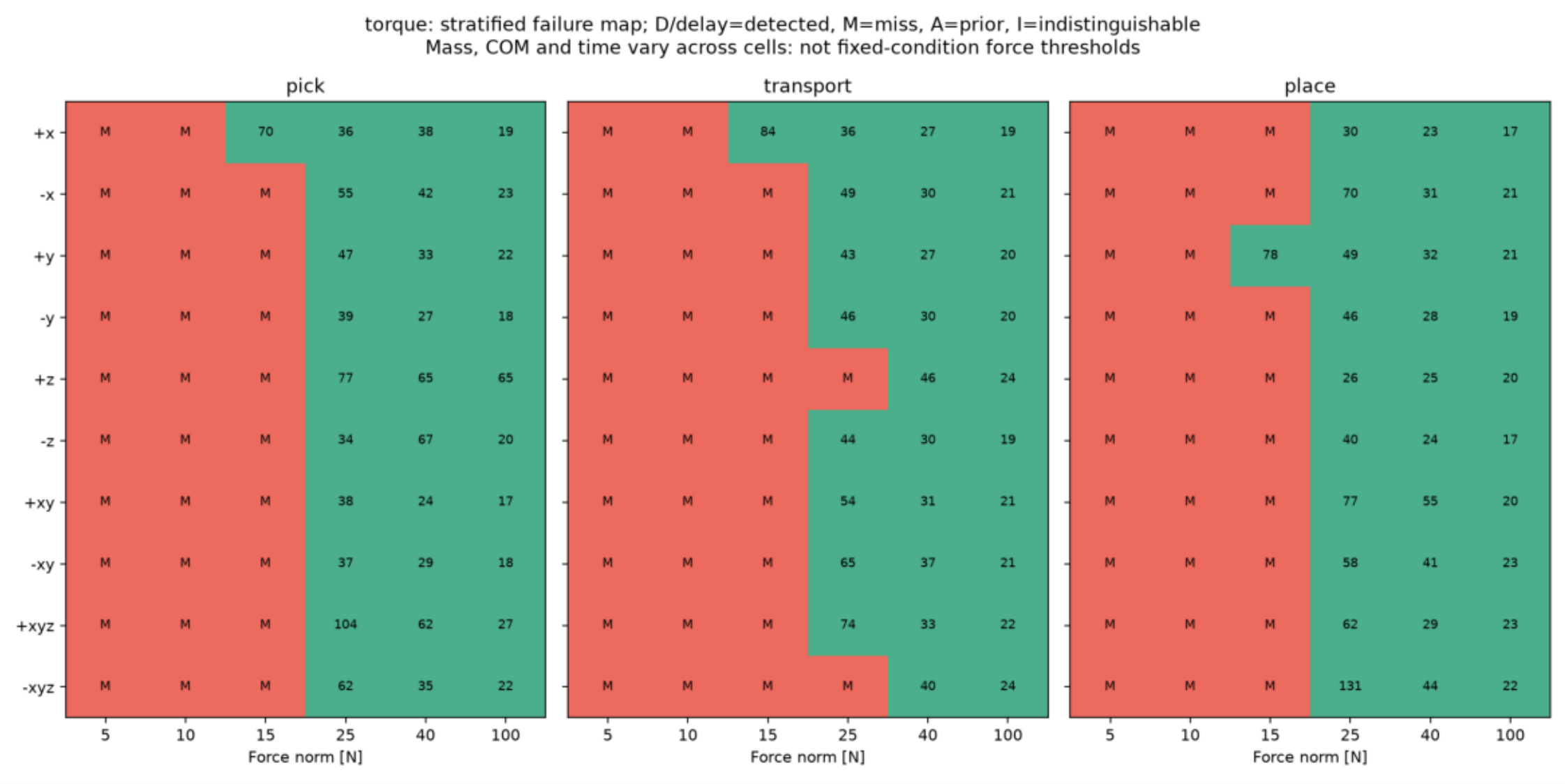
Collision detection: estimated external force



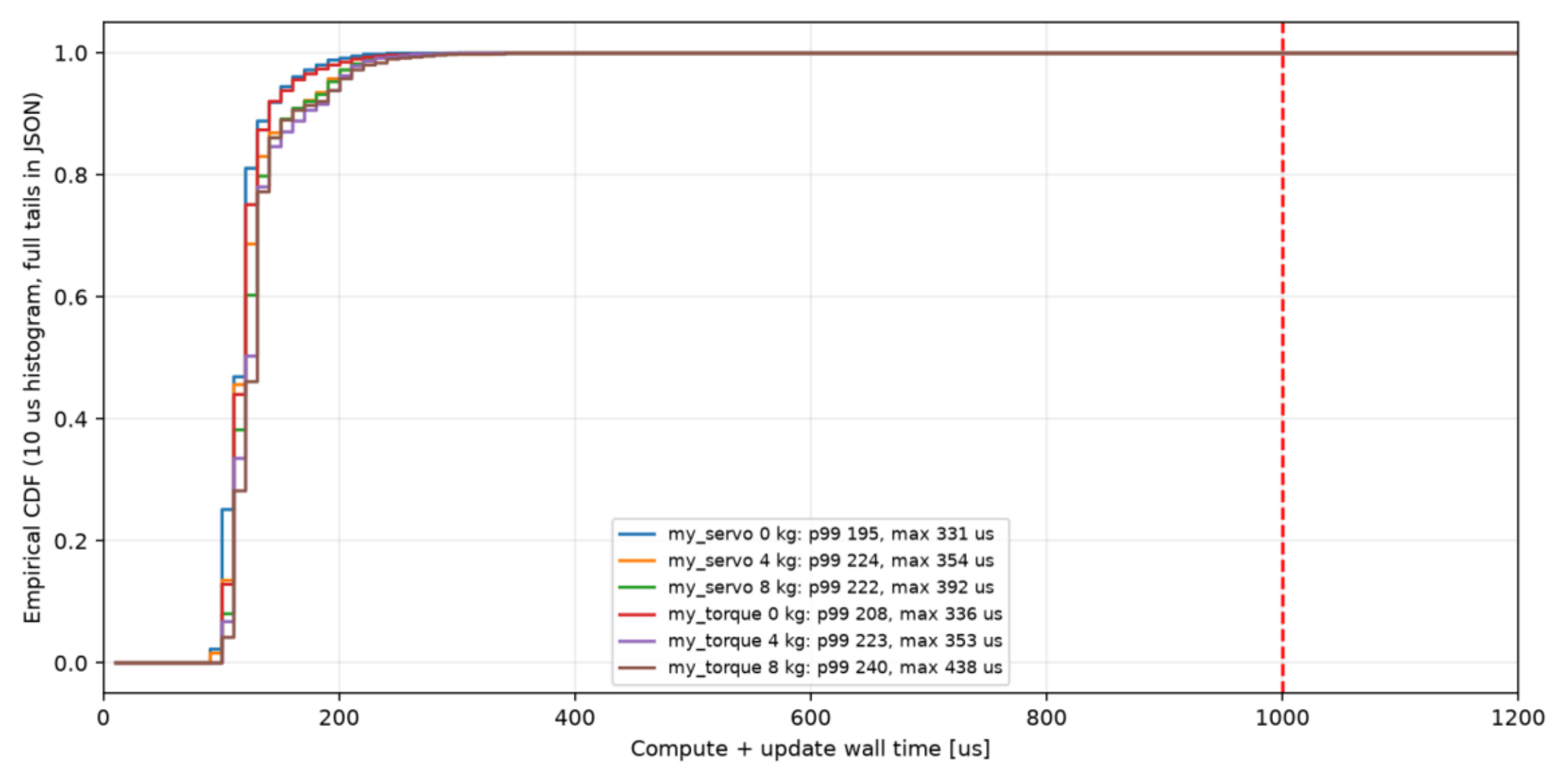
A finite-condition failure map



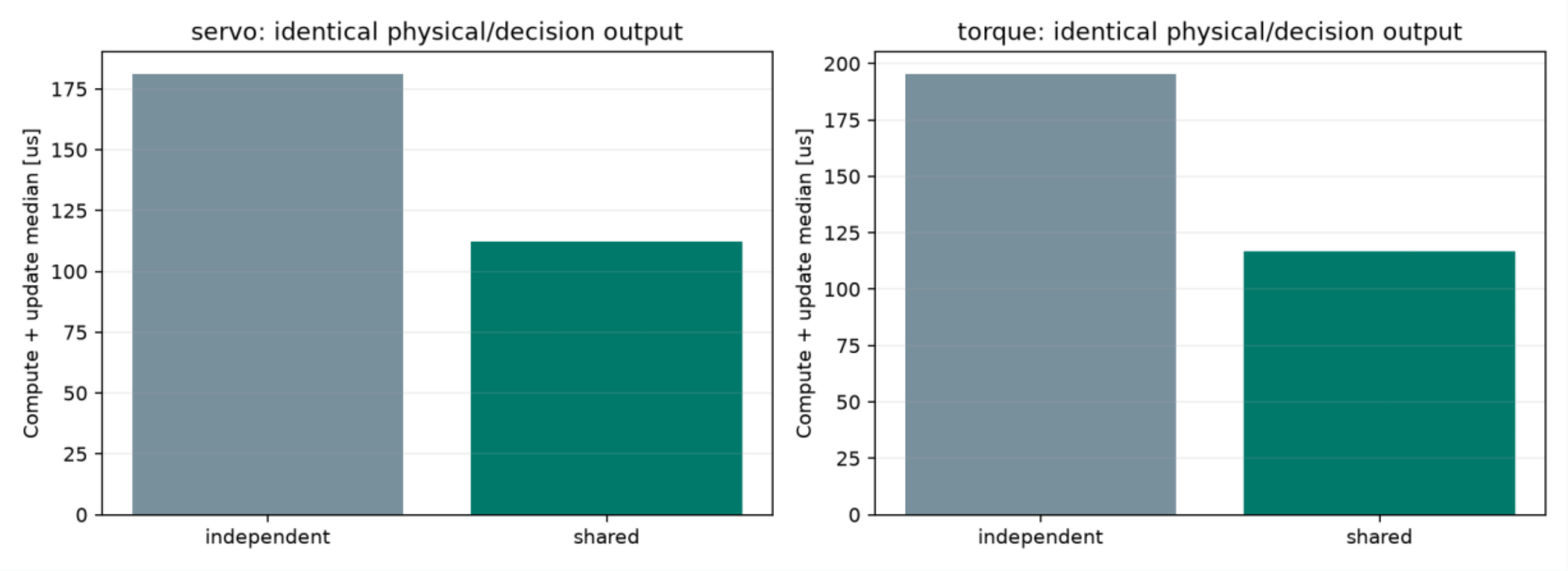
B finite-condition failure map



Computation: distributions, not a hard RT claim



Reuse contribution: numerically identical decisions



Utility, scope and costs

Elastic feedforward is useful independently of transfer estimation; continuous payload FF greatly reduces switching steps.

A gives better tracking and less friction sensitivity here; B gives direct torque authority but more model sensitivity.

770 stratified nominal episodes + 50 analysis-only perturbations are finite coverage, not probabilities.

Vertical force on a feasible table-normal line is indistinguishable from support with these signals.

No hardware validation, collision-triggered stop or safety certification. Additional sensing/process design is needed.

Reproduction and primary references

Use the separately supplied unchanged course kit; our public package excludes its code/models.

README and report tables include commands, software versions and artifact checksums.

De Luca & Mattone, ICRA 2005: generalized-momentum collision residual.

Albu-Schaeffer, Ott & Hirzinger, IJRR 2007: flexible-joint torque feedback and inertia shaping.

MuJoCo official computation documentation; exact links and caveats in docs/REFERENCES.md.